Ali Saeed Chaudhary

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Mechanical Engineering student at Iowa State University with a 4.0 CGPA and hands-on experience in mechanical design, prototyping, and robotics. Leads mechanical subsystem development for a NASA Lunabotics competition team and conducts aircraft trajectory prediction research in collaboration with RTX Collins Aerospace. Proficient in SolidWorks, Python, and additive manufacturing, with experience developing hardware for robotics and rocketry applications.

EDUCATION

Iowa State University — Ames, Iowa
B.S., Mechanical Engineering, Cyber-Physical Systems Minor. ***CGPA: 4.0***. — *May 2028*

Korea University — Seoul, South Korea
Study Abroad Program. — *Feb - Jun 2026*

Relevant Coursework: Mechatronics, Reinforcement Learning with Robotics Applications, Nuclear Power

EXPERIENCE

Research Assistant — Aug 2026 - Present
Iowa State University, Coordinated Systems Lab — *Ames, Iowa*

- Developing RNN-based models to predict deviations from planned aircraft trajectories and estimate forecast uncertainty.
- Completed preprocessing pipelines for the NASA Sherlock database, prepared trajectory-deviation targets for model training.
- Validated with test data and measured flight deviation within 100ft and 1 knot tolerances

Radiology IT Support Coordinator — Nov 2024 - Feb 2026
Concord Medical Management, LLC — *Flower Mound, Texas (remote)*

- Acted as primary point of contact for radiologists and healthcare facilities, diagnosing and resolving time-sensitive technical issues affecting medical imaging workflows across hundreds of facilities throughout the U.S.
- Owned troubleshooting and issue escalation, coordinating directly with facilities to restore system functionality.

Lab Engineering Intern — May 2024 - Aug 2024
LUMS FMOD Research Lab — *Lahore, Pakistan*

- Led design and construction of a compact substrate rubbing machine with a custom-milled aluminum frame, Arduino Mega/RAMPS 1.4 control system driving three NEMA 17 stepper motors.
- Designed the mechanical assembly in Onshape, developed a custom control panel PCB, and 3D-printed control surfaces and button caps for automated operation and manual adjustment.
- Implemented an automated rubbing routine with roller retraction and limit switch based stage homing; deployed the completed machine to the lab.
- Built a PID-controlled temperature-regulated microscope stage as a secondary project during the same internship.

ACTIVITIES & LEADERSHIP

Cardinal Space Mining, NASA Lunabotics — Aug 2026 - Present
Collections Sub-Team Member

- Leading dust cover design for the collections sub-system of a lunar regolith mining rover
- Designing to improve robustness over the prior year's design while maintaining functionality and manufacturability.

Cyclone Rocketry, sponsored by Boeing — Oct 2024 - Dec 2025
Mechanical Sub-Team Member

- Proposed and led design of a rocket airbrake mechanism which was installed and launched in the club’s rocket
- Modeled components in SolidWorks and ran FEA simulations in Ansys.

PROJECTS

RL-Trained Robotic Hand — May 2026 - Jul 2026

- Trained a reinforcement learning policy in Isaac Sim to control a robotic hand using reference data from the HO-Cap (NeurIPS 2025) dataset.
- Engineered a custom reward function prioritizing fingertip and palm contact, stable grasp formation, and reduced unnecessary motion.
- The trained model successfully grabbed and manipulated objects in Isaac Sim.

Private Home Server — Jun 2026 - Present

- Building a private media server for storing, hosting, and distributing services and files.
- Configured remote access through an Oracle VPS using Pangolin and Traefik as a reverse proxy, routing multiple services behind a single entry point.

SKILLS

CAD & Simulation: SolidWorks, AutoCAD, Onshape, Fusion 360, Isaac Sim, Ansys FEA, CAM
Manufacturing: CNC Milling, Welding (TIG, MIG), Metal Working, 3D-Printing
Programming & ML: Python, RNNs, PPO, SAC, VLM / VLA Models, Robot OS, Arduino, STM32, PID Control